Part F:

Question: How useful was ChatGPT for reviewing the code and identifying possible issues?

Answer: ChatGPT was very useful for reviewing the C++ code because it helped identify invalid inputs, edge cases, and possible improvements. It pointed out that negative values, non-numeric input, floating-point values, and very large integers could cause problems. It also suggested using input validation such as cin.fail() and checking whether daysLate is negative. This made the code review more systematic.

Question: How useful was GitHub Copilot for debugging the program inside VS Code?

Answer: GitHub Copilot was useful for debugging because it worked directly inside VS Code and helped identify problems while working with the program. It could provide suggestions based on the code and help understand possible corrections quickly. I found it helpful for getting immediate coding assistance without switching to another application. However, its suggestions still needed to be checked by running the program and verifying whether the proposed changes actually solved the problem.

Question: How did Perplexity help you verify suggestions and examine supporting sources?

Answer: Perplexity was useful for verifying technical suggestions because it provided explanations along with supporting sources. This made it easier to check whether a suggested programming practice was reliable rather than accepting an AI-generated answer immediately. By examining the referenced sources, I could compare the information with official or trustworthy documentation. This was especially helpful when I wanted evidence behind a technical explanation or wanted to understand a programming concept in more detail.

Question: Which tool would you prefer for code review, debugging and technical research? Give a reason for each choice.

Answer: For code review, I would prefer ChatGPT because it can systematically analyze code, identify edge cases, invalid inputs, and suggest improvements. For debugging, I would prefer GitHub Copilot because it works directly inside VS Code and provides coding assistance while developing. For technical research, I would prefer Perplexity because it provides supporting sources that can be examined to verify the information.

Question: Why should a software developer verify AI-generated suggestions before using them?

Answer: A software developer should verify AI-generated suggestions because AI can sometimes provide incorrect, incomplete, or unsuitable solutions. A suggestion may appear

reasonable but contain logical errors, introduce new bugs, or fail for certain inputs and edge cases. In this program, for example, input validation is important because negative, non-numeric, floating-point, or extremely large values can cause issues. Testing and checking reliable documentation helps ensure the final code is correct and safe.